Trideep Bishal Bhowmik

Chemical Engineering Undergraduate

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Education

Jadavpur University,Kolkata **2024-2028**
B.E. in Chemical Engineering *CGPA: 8.08 / 10*

Relevant Coursework

- Heat Transfer
- Transport Phenomena
- Thermodynamics
- Process Calculations
- Mechanical Operations
- Reaction Engineering
- Fluid Mechanics
- Mass Transfer

Experience

Research Intern **June 2026 – July 2026**
Department of Chemical Engineering, IIT Guwahati *Supervisor: Dr. Siddhartha Mukherjee* *Tools Used: MATLAB*

- Conducted research on **Pressure-Driven Blood Flow in Deformable Microchannels: Impact of the Fåhræus–Lindqvist Effect** under the guidance of **Dr. Siddhartha Mukherjee**.
- Developed a mathematical model incorporating the **Carreau non-Newtonian fluid model**, hematocrit-dependent rheology, and the Fåhræus–Lindqvist effect to analyze blood flow in microchannels.
- Applied concepts of **solid mechanics** to model deformable microchannel walls and investigate fluid–structure interaction in biomicrofluidic systems.

Projects

Biogas Production and Process Analysis ***Neptune Laboratory, Jadavpur University***

- Conducting a research project under the guidance of **Prof. Achintya Mukhopadhyay**, focusing on biogas production through anaerobic digestion and process performance analysis.
- Investigating the influence of key operating parameters on biogas yield, gas generation, and overall process efficiency.
- Analyzing experimental data and process characteristics to understand the factors governing anaerobic digestion and biogas production.

Skills

Programming & Scripting: C, Python, SQL, MATLAB, LATEX
Engineering Software: AutoCAD, Advanced MS Excel
Data Visualization: Power BI, Tableau
Analytical Skills: Critical Thinking, Problem Solving, Numerical Methods

Industrial Exposure

Industrial Visit – IVL Dhunseri Plant **2026**
Haldia, West Bengal

- Gained practical exposure to **industrial plant operations**, process equipment, and large-scale chemical manufacturing practices.
- Observed **process flow, safety practices, utilities**, and operational procedures followed in an industrial environment.

Certifications

- **Introduction to Tech Entrepreneurship – IIT Bombay:** Successfully completed a course focused on startup fundamentals, innovation, and technology-driven entrepreneurship.
- **Personal and Corporate Finance Workshop – Srijan, Jadavpur University, Kolkata:** Gained insights into financial planning, investment concepts, and corporate finance basics.
- **Advanced MS Excel Workshop – Azeotropy, IIT Bombay:** Developed practical skills in data analysis, spreadsheet modelling, and advanced Excel functions.

Position of Responsibility

Event Management Coordinator **Feb 2026 – Present**
Chemical Engineering Learning League (ChELL), Jadavpur University

- Organized and coordinated **interactive sessions** with alumni and industry professionals to facilitate knowledge and experience sharing.
- Managed **career-oriented sessions** focused on placement, internship opportunities, and professional development.